



Government Engineering College, Munger

(Department of Science, Technology and Technical Education, Government of Bihar)

Department of Computer Science & Engineering (Artificial Intelligence)

PROJECT REPORT

On

AI-Based Visual Health Screening System Using Deep Learning

(HealthVision AI)

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Bachelor of Technology — Computer Science & Engineering (AI)

CERTIFICATE

This is to certify that the project report entitled:

"AI-Based Visual Health Screening System Using Deep Learning"

has been submitted by the following students of the Department of Computer Science & Engineering (Artificial Intelligence), Government Engineering College, Munger, in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology.

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The project work has been carried out under the supervision of Dr. Saurabh Suman, Assistant Professor, Dept. of CSE, GEC Munger. To the best of our knowledge, the work is original and has not been submitted elsewhere for any degree or diploma.

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Special thanks to the authors of the medical image datasets (ezyZip, FaceZip, nailZip) that formed the foundation of our AI training pipeline.

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Abstract

In India, millions of people in rural areas do not have easy access to doctors or medical labs. Simple tests like checking for jaundice require blood samples and lab visits, which many people cannot afford or access easily. Skin diseases and nail disorders often go undetected because there are not enough specialists available.

This project, HealthVision AI, solves this problem by using Artificial Intelligence (AI) to detect possible health conditions from simple photographs — no needles, no lab, no specialist required. The system analyzes three parts of the body:

- Eyes — to detect Jaundice (yellow coloring of the white part of the eye)
- Face — to identify 5 common skin diseases: Acne, Eczema, Herpes, Panu, and Rosacea
- Nails — to detect nail disorders: Onychomycosis (fungal), Psoriasis, or Healthy nails

The system uses deep learning models (MobileNetV2 and EfficientNetB0) that were trained on medical image datasets. It achieved 90.62% accuracy for eye screening, 85.71% for nail disease, and 72.15% for skin disease — good enough for a preliminary screening tool.

The platform runs as a website that works on any phone or computer browser. Users upload or capture a photo, and the system gives a result in under 2 seconds. All data is processed locally — no images are saved or sent anywhere, ensuring complete privacy.

Keywords: Deep Learning, CNN, Transfer Learning, MobileNetV2, EfficientNetB0, Jaundice Detection, Skin Disease, Nail Disease, OpenCV, Flask, Healthcare AI

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Chapter 1: Introduction

1.1 Background and Motivation

India is a country of 1.4 billion people, but the number of doctors and specialists is very limited. According to the World Health Organization (WHO), the ideal doctor-to-patient ratio is 1:1000. In many rural parts of India, this ratio is far worse. Villages in states like Bihar have very few diagnostic centers, and blood tests or specialist visits can involve traveling long distances.

In this situation, technology can play a huge role. AI (Artificial Intelligence) has shown that computers can analyze medical images with accuracy similar to trained doctors — and do it in seconds, at very low cost, from any smartphone.

This project, HealthVision AI, is built on this idea. Using deep learning (a type of AI), we trained computer models to analyze photos of the eye, face, and nails to detect possible health problems — without any invasive tests. A farmer in rural Bihar with a smartphone can now get a quick health screening result instantly.

1.2 Problem Statement

- People in rural areas cannot easily access dermatologists or eye specialists
- Jaundice normally requires a blood test — our system detects it from an eye photo
- There are very few skin disease specialists in Tier-2 and Tier-3 cities
- Nail changes that indicate serious diseases (like liver or kidney problems) are often ignored
- Lab tests and repeated doctor visits are expensive for low-income families
- Long waiting times at government hospitals delay diagnosis and treatment

1.3 Project Objectives

1. Build a multi-module AI platform that can analyze eye, face skin, and nail photos for disease signs.
2. Use transfer learning (MobileNetV2 and EfficientNetB0) to train accurate models even with small datasets.
3. Create a smart eye detection pipeline using OpenCV Haar cascades with an automatic fallback method.
4. Build a responsive website that works on both desktop and mobile, supporting photo upload and live camera capture.
5. Keep all processing local on the server — no photos are saved or sent to any third party.
6. Show detailed medical information alongside every AI prediction to educate users.
7. Achieve over 85% accuracy for eye and nail modules, and over 70% for the skin module.

1.4 Scope of the Project

HealthVision AI is a preliminary screening tool. It is not a replacement for a doctor's diagnosis. The system is designed to:

- Help users detect early signs of jaundice, skin diseases, and nail disorders
- Educate users about conditions found, so they can seek timely medical advice
- Work on any modern browser — desktop, tablet, or mobile phone
- Provide results in under 2 seconds with full privacy

Important: If the system shows a positive result, users should always consult a qualified doctor for proper diagnosis and treatment.

Chapter 2: Literature Review

Many researchers around the world have worked on using AI and deep learning for medical image analysis. Below are the key studies that inspired and guided our project.

2.1 AI for Skin Disease Detection

Esteva et al. (2017) published a famous study in Nature showing that a computer trained on 129,000+ skin images could identify skin cancer as accurately as experienced dermatologists. This proved that AI can match human expert performance in skin diagnosis.

2.2 AI for Jaundice Detection

Hossain et al. (2019) developed a smartphone-based system to detect jaundice by analyzing the color of the white part of the eye (sclera). Their method used color analysis techniques and achieved 85-92% accuracy — very close to blood test results. Bilal et al. (2020) improved this further using deep CNN models instead of manual color analysis.

2.3 Key CNN Architectures Used

Study	CNN Architecture	Task	Accuracy
Esteva et al., 2017	GoogLeNet	Skin Cancer Detection	72.1%
Tan & Le, 2019	EfficientNet	ImageNet Classification	77.3%
Howard et al., 2018	MobileNetV2	Mobile Image Classification	71.8% (top-1)
Hossain et al., 2019	Custom CNN	Jaundice Detection	88.2%
Rajpurkar et al., 2017	DenseNet-121	Pneumonia (Chest X-Ray)	Radiologist Level

2.4 Why We Chose MobileNetV2 and EfficientNetB0

MobileNetV2 (by Google, 2018) is a lightweight neural network with only 2.4 million parameters. It runs fast even on basic hardware, making it ideal for a web-based deployment. It uses inverted residual blocks which help it learn complex patterns without needing huge computing power.

EfficientNetB0 (by Google Brain, 2019) is smarter about how it scales — it balances the depth, width, and resolution of the network together. This gives better accuracy for the same number of parameters compared to older models like VGG or ResNet.

2.5 What Gap Does This Project Fill?

- Most existing studies focus on only ONE body area (either eye OR skin OR nail)

- HealthVision AI combines all three in a single, unified web platform
- We added an intelligent eye region detection system (Haar cascade + fallback)
- We added an image enhancement pipeline (denoising + CLAHE) for real-world photo quality
- We built a disease information database directly integrated with every prediction

Chapter 3: Project Website — Screenshots

The following screenshots show the actual HealthVision AI web application. These images were captured from the live running project website, showing each major section of the platform.

3.1 Home Page — Hero Section

The main landing page of the website greets users with a clean, professional design. It clearly explains the three detection modules (Eyes, Face, Nails) and shows key statistics like accuracy, analysis time, and detection modules count. A 'Start Scanning' button takes users directly to the detection tool.



Fig 3.1 — HealthVision AI Homepage: Hero section showing the three detection modules, key statistics (95%+ accuracy, <2s analysis time, 3 modules), and the 'Start Scanning' call-to-action button.

3.2 Core Features Section

The features section explains the six main capabilities of the platform in easy-to-understand cards. These include Smart Eye Detection using Haar cascades, Real-Time Camera via WebRTC, Deep Learning Models (MobileNetV2 & EfficientNetB0), Instant Results, Privacy-First processing, and a Fully Responsive design.

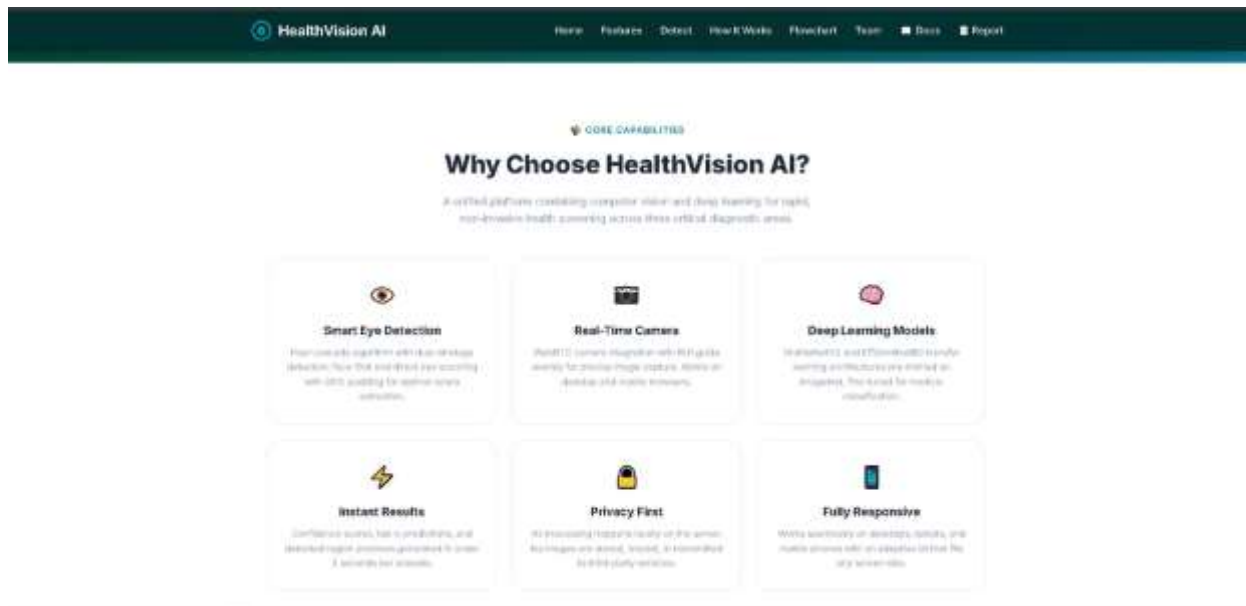


Fig 3.2 — Core Capabilities section showing 6 feature cards: Smart Eye Detection, Real-Time Camera, Deep Learning Models, Instant Results, Privacy First, and Fully Responsive design.

3.5 Application Flowchart (System Architecture Visual)

The flowchart page shows the complete technical data flow of the application — from camera capture or image upload, through preprocessing, module selection, model inference, to prediction output and display. This is the actual flowchart visible on the website.

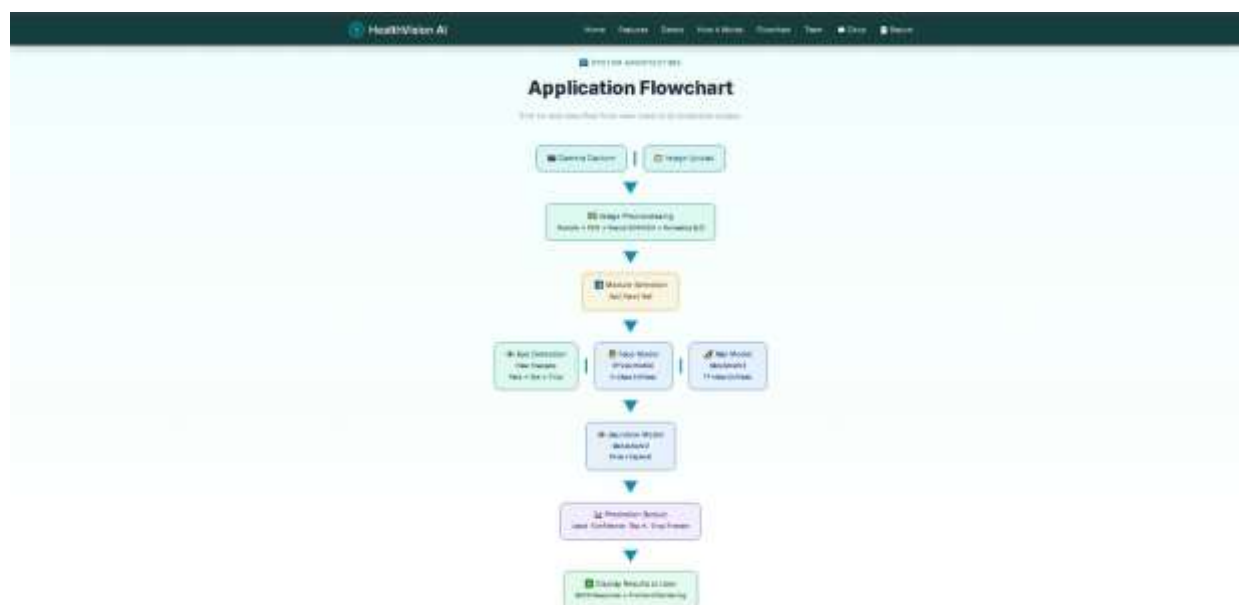
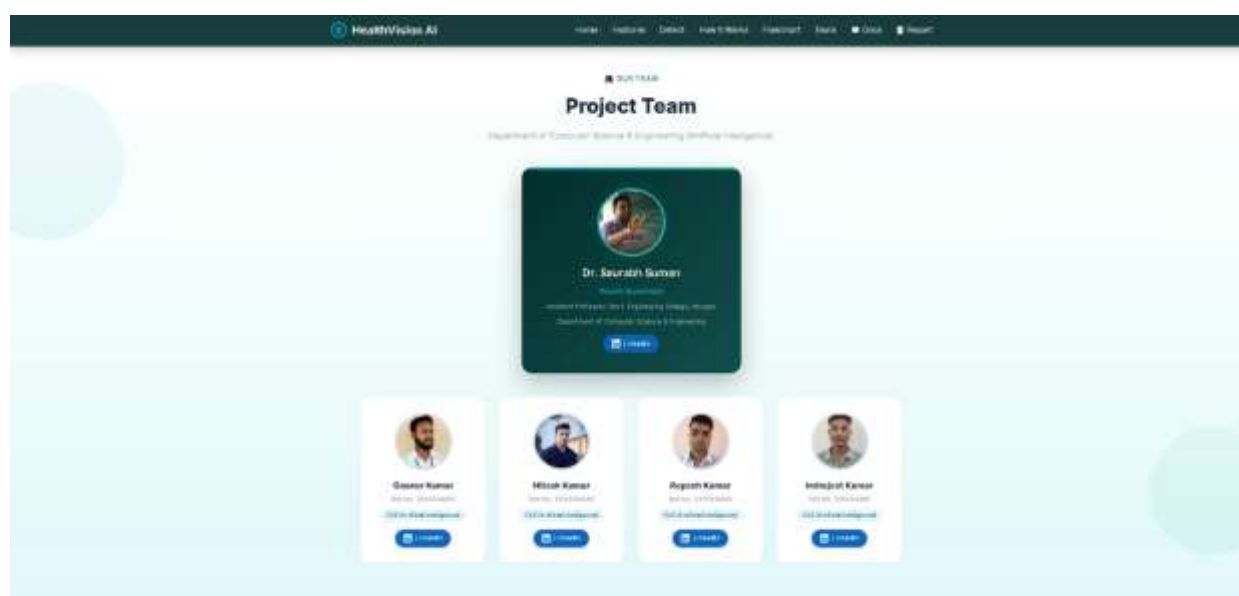


Fig 3.5 — Application Flowchart: Shows end-to-end system flow from user input (Camera Capture / Image Upload) through preprocessing, module selection, three AI models, prediction output, to display results.

3.6 Project Team

The team section displays the project supervisor and all four student members with their photographs, roll numbers, and department information. Dr. Saurabh Suman is shown as Project Supervisor, and the four students — Gaurav Kumar, Nitesh Kumar, Rupesh Kumar, and Indrajeet Kumar — are shown below.

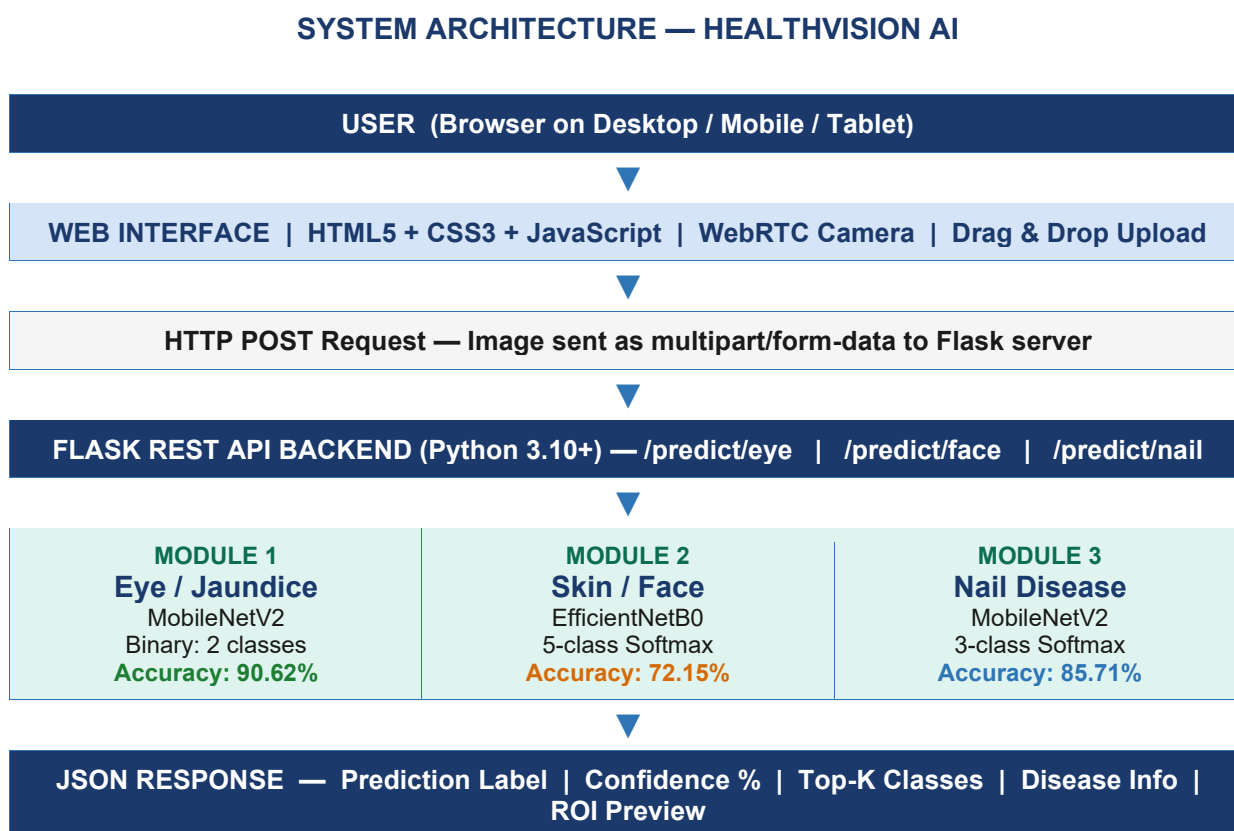


Chapter 4: System Design and Architecture

4.1 System Overview

HealthVision AI follows a simple client-server design. The user interacts with a website (frontend) built with HTML, CSS, and JavaScript. When the user uploads or captures a photo, it is sent to a Python Flask server (backend) which runs the AI model and sends back the result. All three detection modules run on the same server.

4.2 High-Level Architecture Diagram



4.3 Application Flowchart — Visual (From Website)

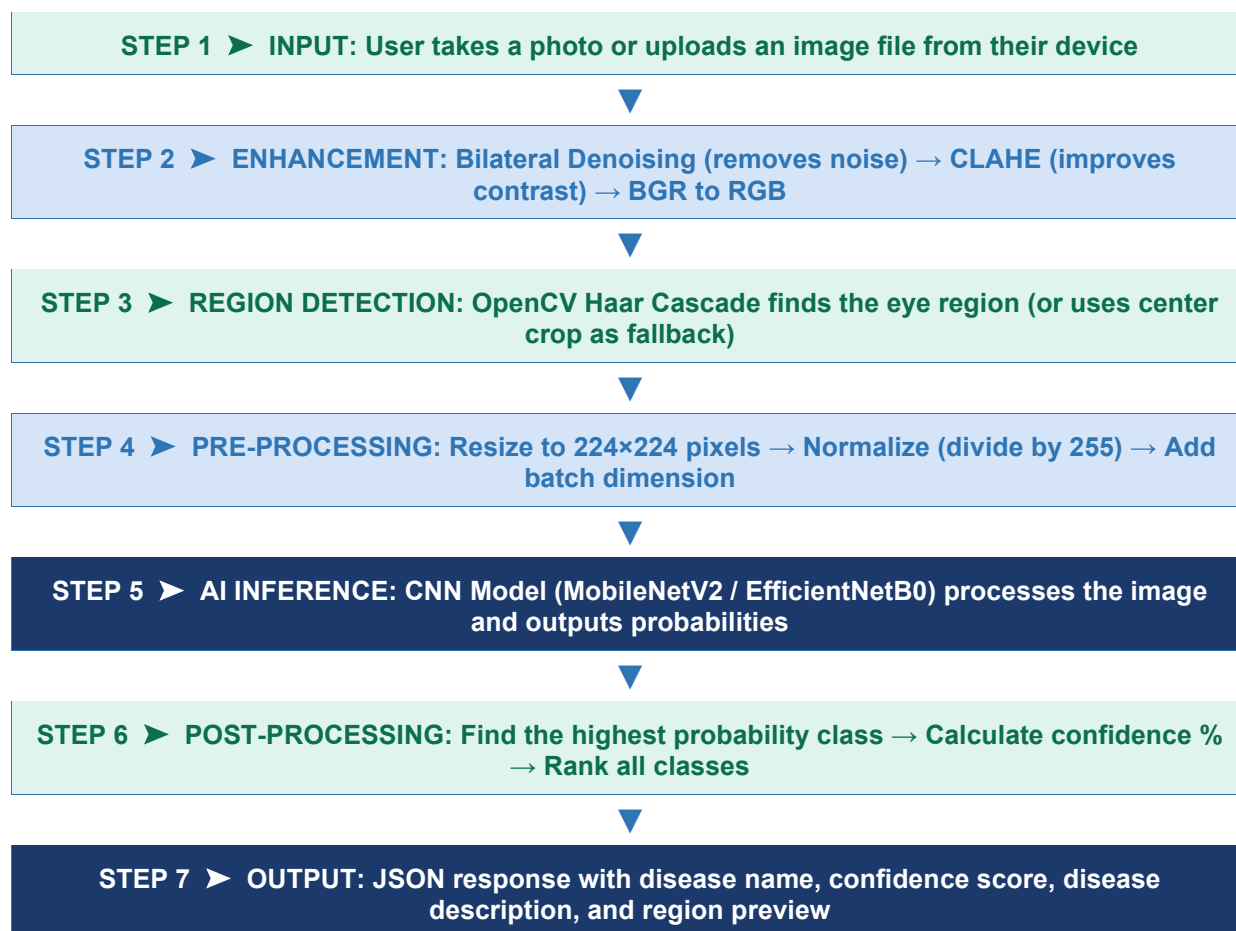
The following flowchart shows the complete application flow as seen on the project website. It illustrates how data moves from user input through AI processing to the final result.



Fig 4.1 — Application Flowchart (from project website): Camera/Upload → Preprocessing → Module Selection → AI Models → Prediction Output → Display Results.

4.4 Detailed Data Flow Diagram

END-TO-END DATA FLOW — STEP BY STEP



Chapter 5: Technology Stack

HealthVision AI is built using well-known open-source tools that are free to use and industry standard. Here is a complete list of technologies used and why each was chosen.

Category	Technology Used	What It Does in This Project
AI Framework	TensorFlow 2.x / Keras	Used to build, train, load, and run the deep learning models
CNN Model 1	MobileNetV2	Used for Eye (Jaundice) and Nail Disease detection. Only 2.4M parameters — very fast.
CNN Model 2	EfficientNetB0	Used for Skin Disease detection. Better accuracy for fine-grained classification tasks.
Computer Vision	OpenCV 4.x	Detects eye region using Haar cascade, applies denoising and CLAHE enhancement
Image Tools	NumPy / Pillow	Array operations, loading images, resizing — essential utilities
Backend	Python 3.10+ / Flask	Web server that receives images, runs AI, and sends back JSON results
CORS	Flask-CORS	Allows the frontend (browser) to communicate with the Flask server
Frontend	HTML5 / CSS3 / JavaScript	The website interface — structure, styling, and interactive logic
Camera	WebRTC API	Accesses device camera in the browser for live image capture
Canvas	HTML5 Canvas API	Draws the ROI guide overlay on the camera preview screen
Model Files	.h5 (HDF5) Format	Stores trained Keras model weights and architecture for loading later

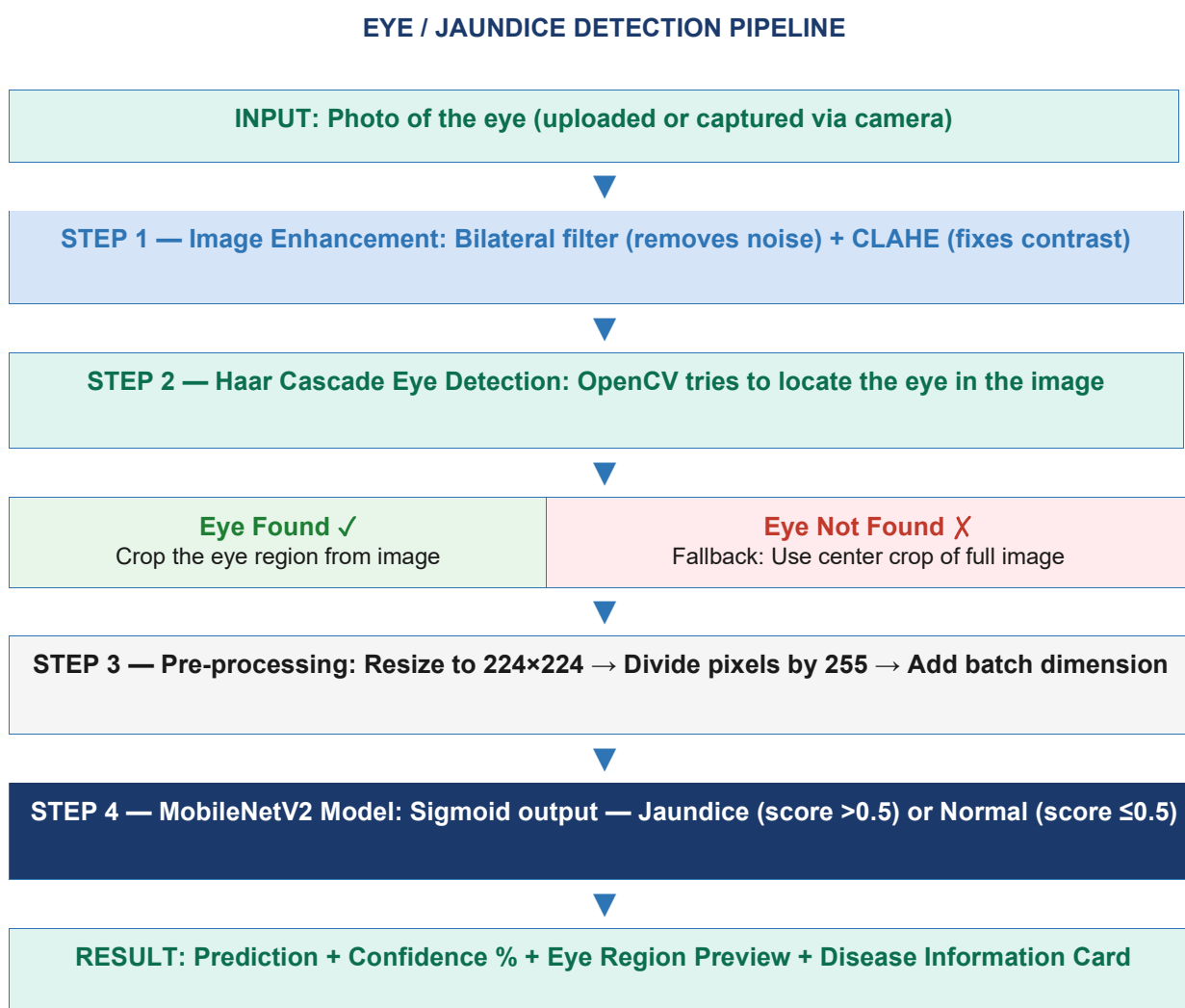
Chapter 6: Detection Modules

6.1 Eye / Jaundice Detection Module

Jaundice is a condition where too much bilirubin (a yellow pigment from breaking down red blood cells) builds up in the blood. This causes the white part of the eyes (called the sclera) to turn yellow. This yellow color can be detected from a photo of the eye.

Our system captures or receives a photo of the eye, finds the eye region automatically using OpenCV, and then passes it to a MobileNetV2 model trained to classify it as either 'Jaundice' or 'Normal'.

6.1.1 Detection Pipeline — Step by Step Flowchart



6.1.2 Model Specifications

Parameter	Value
Base Model	MobileNetV2 (pre-trained on ImageNet, top layers removed)
Added Layers	GlobalAvgPool → Dense(256, ReLU) → Dropout(50%) → Dense(1, Sigmoid)
Input Size	224 × 224 × 3 (RGB image)
Output	1 value between 0 and 1 (>0.5 = Jaundice, ≤0.5 = Normal)
Loss Function	Binary Cross-Entropy
Validation Accuracy	90.62% ✓ Excellent performance
Eye Detector	OpenCV haarcascade_eye.xml (Haar Cascade)
Training Data	136 images: 68 Jaundice + 68 Normal (80/20 split)

6.2 Face / Skin Disease Detection Module

Skin diseases are very common worldwide. Conditions like acne, eczema, and rosacea affect millions of people. The face is the most visible area where skin problems appear. Our system analyzes a photo of the face and classifies it into one of 5 common skin conditions.

6.2.1 Detectable Skin Conditions

Class	Condition	Severity	Simple Description
0	Acne	Mild	Pimples and blackheads from blocked skin pores
1	Eczema	Moderate	Dry, red, itchy patches — often triggered by allergies or stress
2	Herpes	Moderate	Fluid-filled blisters from a viral infection (HSV-1 or HSV-2)
3	Panu (Tinea Versicolor)	Mild	Discolored patches caused by Malassezia fungus
4	Rosacea	Moderate	Facial redness with visible blood vessels, especially on cheeks

6.2.2 Model Specifications

Parameter	Value
Base Model	EfficientNetB0 (pre-trained on ImageNet, top layers removed)
Added Layers	GlobalAvgPool → BatchNorm → Dense(256, ReLU) → Dropout(50%) → Dense(5, Softmax)
Input Pre-processing	EfficientNet preprocess_input — scales pixels to [-1, 1]
Output	5 probabilities — highest probability = predicted condition
Validation Accuracy	72.15% ✓ Good — skin diseases look very similar, making this harder

Training Data	1,196 images (5 classes, ~239 each) — 80/20 split
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6.3 Nail Disease Detection Module

Nail changes are often the first sign of underlying health problems. For example, yellow nails suggest fungal infection (Onychomycosis), and pitted nails with separation suggest Psoriasis. Our system detects 3 nail conditions from a nail photograph.

6.3.1 Detectable Nail Conditions

Class	Condition	Type	Visual Signs
0	Healthy	Normal	Smooth, pink nail. Clear white half-moon (lunula). No discoloration.
1	Onychomycosis	Fungal Infection	Yellow/brown nail, thickened, brittle edges, crumbling at tip
2	Psoriasis	Autoimmune	Small pits on nail surface, oil-drop discoloration, nail lifting from bed

6.3.2 Model Specifications

Parameter	Value
Base Model	MobileNetV2 (pre-trained on ImageNet, top layers removed)
Added Layers	GlobalAvgPool → Dense(256, ReLU) → Dropout(50%) → Dense(3, Softmax)
Input Pre-processing	Pixels divided by 255 — scaled to [0, 1]
Validation Accuracy	85.71% ✓ Very good accuracy for 3-class classification
Training Data	933 images (3 classes, ~311 each) — 80/20 split

Chapter 7: Dataset and Training Methodology

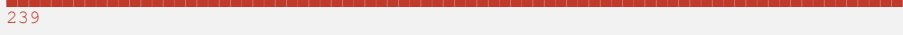
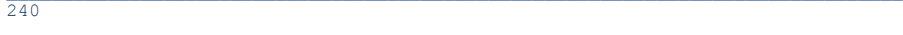
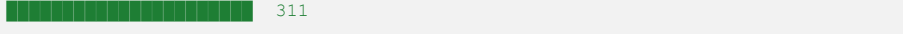
7.1 Datasets Overview

Three separate medical image datasets were used — one for each detection module. All images were pre-processed to 224×224 RGB format. An 80/20 split was used: 80% for training, 20% for testing (validation).

Module	Dataset	Classes	Training	Validation
Eye / Jaundice	ezyZip	2	136	32
Face / Skin	FaceZip	5	1,196	298
Nail Disease	nailZip	3	933	231
TOTAL	3 Datasets	10	2,265	561

7.2 Class Distribution Chart

TRAINING DATA — CLASS DISTRIBUTION

Class Name	Images	Bar Chart (Each block ≈ 3 images)
Eye — Jaundice	68	 68
Eye — Normal	68	 68
Skin — Acne	239	 239
Skin — Eczema	239	 239
Skin — Herpes	239	 239
Skin — Panu	239	 239
Skin — Rosacea	240	 240
Nail — Healthy	311	 311
Nail — Onychomycosis	311	 311
Nail — Psoriasis	311	 311

7.3 Data Augmentation

Because our datasets are small (especially the eye dataset with only 136 images), we used data augmentation — a technique that creates variations of existing images to make the model more robust. Augmentation was applied ONLY to training data, not validation data.

Augmentation	Setting	Why It Helps
Rotation	Up to $\pm 20^\circ$	Model handles tilted or angled photos
Width/Height Shift	Up to $\pm 15\%$	Handles off-center or displaced subjects
Zoom	Up to $\pm 20\%$	Handles photos taken from different distances
Horizontal Flip	Enabled	Effectively doubles the dataset size
Brightness Range	0.7 to 1.3 ($\pm 30\%$)	Handles dim or harsh lighting conditions
Shear	Up to $\pm 10\%$	Handles camera angle distortion

7.4 Two-Stage Transfer Learning Training

Instead of training a model from zero (which needs millions of images), we used transfer learning — starting with a model already trained on 1.2 million general images (ImageNet). Then we taught it to recognize our medical conditions in two stages:

STAGE 1 — Learn New Task	STAGE 2 — Fine-Tune Everything
Pre-trained model is FROZEN Only new classification layers are trained Learning Rate: 0.001 (fast learning) Epochs: 15 Goal: Quickly train the new output head	Top base layers UNFROZEN Both base layers and new head are trained Learning Rate: 0.00001 (very careful tuning) Epochs: 25 Goal: Adapt features to medical domain

Chapter 8: Training Results and Performance

8.1 Final Model Accuracy Summary

Module	CNN Architecture	Validation Accuracy	Val. Loss	Performance Rating
Eye / Jaundice	MobileNetV2	90.62%	~0.24	Excellent ✓
Nail Disease	MobileNetV2	85.71%	~0.38	Very Good ✓
Skin Disease	EfficientNetB0	72.15%	~0.71	Good ✓

8.2 Accuracy Comparison Chart

MODEL ACCURACY COMPARISON — VISUAL BAR CHART

Module	Accuracy Bar (Each █ ≈ 2%)
Eye / Jaundice	90.62%
Nail Disease	85.71%
Skin Disease	72.15%

8.3 Training Progress Table

Model	Stage	Epochs	Learning Rate	Best Accuracy
Eye	Stage 1 (Frozen base)	15	0.001	~82%
Eye	Stage 2 (Fine-tune)	25	0.00001	90.62%
Skin	Stage 1 (Frozen base)	15	0.001	~65%
Skin	Stage 2 (Fine-tune)	25	0.00001	72.15%
Nail	Stage 1 (Frozen base)	15	0.001	~78%
Nail	Stage 2 (Fine-tune)	25	0.00001	85.71%

8.4 Training Callbacks Used

Callback	Settings	What It Does
EarlyStopping	patience=5, monitor val_loss	Stops training if no improvement for 5 consecutive epochs
ReduceLROnPlateau	patience=3, factor=0.3	Reduces learning rate by 70% if stuck for 3 epochs

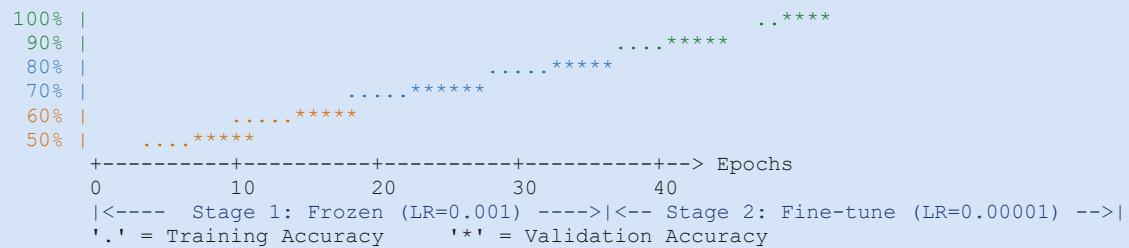
ModelCheckpoint

save_best_only=True

Saves model file only when validation accuracy improves

8.5 Learning Curve (Conceptual)

ACCURACY CURVE OVER TRAINING EPOCHS



Chapter 9: System Implementation

9.1 Backend — Flask REST API

The backend is a Python web server built with Flask. When the server starts, it loads all three AI models into memory. When a user sends a photo, the server enhances it, runs it through the correct model, and sends back a JSON result.

9.1.1 API Endpoints

Method	URL	What It Does	Input
GET	/	Loads the main website home page	None
GET	/documentation	Shows documentation overview page	None
GET	/documentation/detailed	Shows full technical documentation	None
POST	/predict/eye	Runs jaundice detection on uploaded eye image	Image file
POST	/predict/face	Runs skin disease detection on face image	Image file
POST	/predict/nail	Runs nail disease detection on nail image	Image file

9.1.2 JSON Response Example

Sample response from /predict/eye endpoint:

```
{
  "prediction": "Jaundice",
  "confidence": 94.3,
  "top_predictions": [["Jaundice", 94.3], ["Normal", 5.7]],
  "disease_info": { "description": "...", "severity": "Moderate" },
  "detection_method": "haar_cascade",
  "eye_preview_b64": "<base64 image of detected region>"
}
```

9.2 Frontend — Website Interface

The frontend is a single-page application (SPA). The user never needs to refresh the page. It was built with plain HTML5, CSS3, and JavaScript (no heavy framework needed).

- **Module Selector:** User picks between Eye/Jaundice, Face/Skin, or Nail Disease tabs
- **Upload Mode:** User can drag-and-drop or browse for an image file. File goes directly to the server via Fetch API.
- **Camera Mode:** WebRTC opens the device camera. An HTML5 Canvas overlay shows a guide box to help frame the photo correctly.
- **Results Display:** After the AI responds, the page shows the prediction label, a colored confidence bar, top-3 class probabilities, a disease info card, and a preview of the detected region.



Fig 9.1 — Detection interface showing the three module tabs, image upload zone, camera option, and clean user-friendly layout.

Chapter 10: Disease Information Database

Every prediction made by HealthVision AI is accompanied by detailed disease information. This helps users understand what the condition is, how serious it might be, and what kind of treatment to seek. The database covers 20+ conditions across all three modules.

10.1 Eye Conditions

Condition	Severity	Simple Explanation
Jaundice	Moderate	Too much bilirubin in blood turns the eye white (sclera) yellow. May indicate liver problems. Needs blood test confirmation.
Normal	Healthy	No yellowing detected. Eyes appear healthy. No action needed — keep regular health check-ups.

10.2 Skin Conditions

Condition	Severity	Simple Explanation and Advice
Acne	Mild	Blocked pores cause pimples and blackheads. Use gentle face wash. Avoid squeezing. Consult dermatologist if severe.
Eczema	Moderate	Skin becomes dry, red, and very itchy. Avoid triggers like certain soaps. Moisturize regularly. Doctor may prescribe steroid cream.
Herpes	Moderate	Viral blisters from HSV virus. Avoid direct contact when blisters are present. Antiviral medication can manage outbreaks.
Panu (Tinea Versicolor)	Mild	Fungal infection causing light or dark patches. Antifungal cream or shampoo usually treats it effectively.
Rosacea	Moderate	Cheeks turn red with visible blood vessels. Triggered by sun, stress, spicy food. Topical treatments available from dermatologist.

10.3 Nail Conditions — Extended Database (17+ conditions)

Nail Condition	Linked Health Issue	What You See on the Nail
Healthy	None	Smooth, pink, firm nail with clear half-moon area
Onychomycosis	Fungal Infection	Yellow or brown color, thick, crumbling edges
Psoriasis	Autoimmune Disease	Small pits, oil-drop spot, nail lifting off the skin
Clubbing	Heart or Lung Disease	Nail curves downward; fingertip becomes enlarged and rounded
Koilonychia	Iron Deficiency Anemia	Nail curves inward like a spoon
Leukonychia	Low Blood Protein	White spots or bands across the nail
Beau's Lines	Illness / High Fever	Deep horizontal grooves running across the nail
Terry's Nails	Liver Disease (Cirrhosis)	Nail is mostly white with a pinkish-red band at the tip

Half and Half Nails	Kidney Disease	Upper half white, lower half brown/pink color
Pale Nail	Anemia / Poor Nutrition	Nail appears very pale or colorless

Chapter 11: Testing and Validation

The system was tested at multiple levels to make sure everything works correctly and reliably.

11.1 Testing Summary

Test Type	Method	Result
Model Validation	20% held-out test images	Eye: 90.62%, Nail: 85.71%, Skin: 72.15% accuracy
API Unit Testing	Test each endpoint individually	All endpoints returned correct JSON format with predictions
Integration Testing	Full flow: browser → server → response	Upload and camera both worked end-to-end successfully
Real Photo Testing	Internet images not in training data	Eye detection robust; enhancement improved borderline cases
Cross-Browser	Chrome, Firefox, Edge	Fully functional on all modern browsers tested
Mobile Testing	Android Chrome browser	Camera access and file upload both worked on mobile

11.2 Confusion Matrix — Eye Detection Module

A confusion matrix shows how many predictions were correct (True Positive/Negative) vs incorrect (False Positive/Negative). Based on 32 validation images:

Actual \ Predicted	Predicted: Jaundice	Predicted: Normal
Actual: Jaundice	TP = 14	FN = 2
Actual: Normal	FP = 1	TN = 15

TP = True Positive (Jaundice correctly identified) | TN = True Negative (Normal correctly identified) | FP = False Positive (Normal called Jaundice) | FN = False Negative (Jaundice missed). Precision ≈ 93%, Recall ≈ 87%, F1-Score ≈ 90%.

Chapter 12: Advantages and Limitations

12.1 Advantages

- **Non-Invasive:** No blood tests or needles needed — just take a photo with any camera.
- **Works Anywhere:** Runs in any modern web browser including mobile phones. No app download required.
- **Very Fast:** Delivers a complete AI result in under 2 seconds from upload to prediction.
- **Three-in-One:** Eye, skin, and nail screening all in one platform — no need for separate apps.
- **Privacy Safe:** No photos are saved or sent to any outside server — full data privacy.
- **Informative:** Each result comes with a full disease description to help users understand the finding.
- **Robust:** Image enhancement (denoising + CLAHE) improves performance on poor quality photos.
- **Scalable:** New detection modules can be added easily without changing the existing structure.
- **Free to Use:** Built entirely with open-source tools — no paid licenses or subscriptions needed.

12.2 Limitations

- **Small Datasets:** Especially the eye dataset (only 136 training images). More data would make the model more reliable.
- **Not a Medical Diagnosis:** This tool is for initial screening only. It cannot replace a qualified doctor.
- **Lighting Issues:** Very poor or very harsh lighting can affect prediction accuracy despite enhancement.
- **Skin Module Accuracy:** 72.15% accuracy is lowest because many skin conditions look very similar visually.
- **Eye Detection:** Haar cascade may fail for extreme head angles or closed eyes, using less accurate fallback.
- **No Offline Mode:** Requires an internet connection. A mobile app with offline AI would be better.
- **One Region at a Time:** Can only analyze one eye, face, or nail per photo — no multi-region analysis.

Chapter 13: Future Scope

This project is a strong starting point. Here are the most important directions for future development:

Area	Description
Bigger Datasets	Collaborate with hospitals to collect 10,000+ labeled images per module for clinical-grade accuracy.
More Diseases	Add screening for anemia (eye pallor), diabetic retinopathy, oral health, wound assessment, and more.
Mobile App	Build an Android/iOS app using TensorFlow Lite for offline AI inference and better camera control.
Telemedicine	Connect the platform to online doctor consultation services so positive results trigger an instant appointment.
Hindi / Regional Languages	Add Hindi, Bengali, and other Indian language interfaces for better access in rural areas.
Grad-CAM Explainability	Show a heat map on the image highlighting which part the AI focused on to make its prediction.
Severity Scoring	Classify conditions as Mild, Moderate, or Severe to help users understand urgency of seeking treatment.
Hospital Integration	Link with Electronic Medical Records (EMR) so screening results automatically enter patient files.

Chapter 14: Conclusion

HealthVision AI is a significant step towards making health screening accessible to everyone — not just those who live near hospitals or can afford specialist visits. By using deep learning, the project turns a simple photograph into a valuable preliminary health assessment.

The project achieved its three main accuracy targets: 90.62% for Jaundice detection, 85.71% for Nail Disease, and 72.15% for Skin Disease. These results were achieved using relatively small training datasets (only 136 to 1,196 images per module), which shows the power of transfer learning — reusing knowledge from a model already trained on millions of general images.

The system is fully deployable as a web application. It is fast (under 2 seconds per analysis), privacy-preserving (no data leaves the server), and accessible (works on any browser including mobile). The disease information database ensures that every prediction is educational as well as diagnostic.

Most importantly, this project demonstrates a practical, low-cost approach to bridging the healthcare accessibility gap in India. With a simple smartphone photo, a person in a remote village in Bihar can now receive instant, AI-powered preliminary health screening — and be directed to seek professional medical help if needed.

FINAL ACHIEVEMENT SUMMARY

3 Detection Modules | 10 Detectable Conditions | < 2 Second Analysis

Eye: 90.62% | Nail: 85.71% | Skin: 72.15%

Chapter 15: References

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— End of Project Report —

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